

Muhammad Haris

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EDUCATION

Master's in Bioinformatics

Government College University Faisalabad (GCUF)

Faisalabad, Pakistan

Sept. 2025 - Current

CGPA: 3.94/4

Research Focus: Exploring Machine Learning Approaches, Single-Cell Transcriptomics and Advanced Bioinformatics Analyses for Complex Diseases

Bachelor's in Biotechnology

Government College University Faisalabad (GCUF)

Faisalabad, Pakistan

Nov. 2021 - July 2025

CGPA: 3.73/4

Thesis: Identification of Novel Therapeutic Targets for Hypertrophic Cardiomyopathy (HCM) Using RNA-seq Data Analysis

- Identified **6** previously unreported therapeutic targets using Linux and R and implemented pipelines with FastQC, HISAT2, featureCounts, and DESeq2.

RESEARCH EXPERIENCE

Research Assistant

Functional Genomics & Tissue Culture Lab, GCUF

Faisalabad, Pakistan

Sept. 2024 - Current

- Working on Single-cell RNA sequencing analysis to decipher gene expression in individual cells of a particular disease.
- Specialized in maize cell line generation via micropropagation, producing **3** genetically identical lines from a single callus for food safety research.
- Designed and deployed Pakistan's first dedicated Hypertrophic Cardiomyopathy [drug database](#).
- Provided hands-on training to **6** undergraduate students in tissue culture and molecular biology.
- Worked on **6** manuscripts spanning NGS data analysis, tissue culture, database development, and plant biotechnology under climate change.

Summer Research Fellow

International Center for Chemical and Biological Sciences (ICCBS),
University of Karachi

Karachi, Pakistan

June 2025 - July 2025

- Performed DNA library preparation and WES data analysis for Acute Myeloid Leukemia, identifying 884 novel high-impact variants using a custom Linux- and R-based bioinformatics pipeline.
- Conducted variant annotation and pathway enrichment analysis (ANNOVAR, KEGG, and GO) to uncover the functional relevance of identified variants.

Research Intern

Ayub Agricultural Research Institute (AARI)

Faisalabad, Pakistan

July 2024 - August 2024

- Learned molecular biology (DNA extraction, PCR, PAGE, Microscopy) and tissue culture techniques.

Research Intern

Functional Genomics & Tissue Culture Lab, GCUF

Faisalabad, Pakistan

Jan. 2023 - June 2024

- Trained in advanced laboratory techniques, including DNA extraction, PCR/qPCR, PAGE, microscopy, plant tissue culture, microbial isolation, and biochemical enzyme assays.

SKILLS

Biotechnology: Tissue Culture, DNA extraction, qPCR, PCR, Gel Electrophoresis, Bacterial Isolation, Microbiological Techniques, Spectrophotometry, Biochemical Enzyme Analysis, Physiological Plant Health Analysis, Pot Experiment, Plant Extract Preparation

Bioinformatics: scRNA-seq Analysis, Variant Calling, WES & NGS Data Analysis, In-silico Primer Designing, In-silico Vector Designing, Phylogenetic Analysis, Web & Database Development, Drug Repurposing, Functional Enrichment, Pathway analysis, Protein Structure Prediction, Protein-protein Interaction

Languages: LINUX, R, SQL, HTML, CSS, JS, and PHP

Soft Skills: Strong communication, Teamwork, Leadership, Critical Thinking.

PUBLICATIONS

Integrative Transcriptomics and Network Medicine Identified Non-sarcomeric Dysregulated Ion Homeostasis and Innate Immunity genes as Therapeutic Targets in Hypertrophic Cardiomyopathy (HCM)

International Journal of Cardiology (2025) | Under Review

M. Haris, S. Rahman, W. Waheed, M. H. Siddique, F. Azeem*

Perspectives of Plant-Microbes and Nanotechnology against abiotic stress tolerance under climate change conditions

Royal Society of Chemistry (2026) | Under Review

M. Haris, T. Shaheen, S. Zahra, & M. R. Ansari. *

HCMdrugs: A web-based database cataloging the drugs available for the treatment of Hypertrophic Cardiomyopathy (HCM)

Journal of Computational Biophysics and Chemistry (2026) | Published | [DOI](#)

M. Haris, S. Rahman, M. T. Ul Qamar, F. Azeem*

Tissue Culture Optimization: Maize Callus Induction and Factors Affecting the Callus Efficiency

Integrative Plant Biotechnology (2025) | Published | [DOI](#)

M. Haris, S. Rasheed, S. Rahman, I. Fatima, A. Mazhar, F. Azeem *

HONORS & ACHIEVEMENTS

- **Prime Minister's** Letter of Appreciation for Environmental Activism
- **Recognition** at the International Youth Conference 9 (IYC 9), UCLA Luskin (USA)
- **Delegate** - Local Conference of Youth (LCOY) 2024
- **Organized** 25+ events on Environmental Awareness and Conservation as the **President** of Green Youth Movement (GYM) Club, GCUF

TALKS & PRESENTATIONS

- Oral Presentation on “**Sustainable Reuse of Cadmium-Contaminated Textile Wastewater in Wheat Cultivation: A Nano-Physiological and Transcriptomic Perspective**” at BSP Biotechnology Congress - 2025 & “National Dialogue on Biotechnology” at the University of Agriculture, Faisalabad | **16-18th Dec 2025**.
- Talk on “**Importance of Biodiversity for One Earth**” at One-day Seminar on Biodiversity for One Earth: Connecting Civilization through Food, Health, and Energy Resources under Fast Climatic Challenges in Sino-Pak Region, GCUF | **22nd May 2025**
- Presented the **HCMdrugs** database and its future concept at the GCUF Innovation Expo 2025 | **23rd-24th Jan 2025**
- Poster presentation on “**Natural Protein Nanofiber Filters: A Sustainable and Economically Beneficial Solution for Efficient Smog Filtration**” at One-day Symposium on Nanotechnology for Sustainable Environmental Applications, GCUF | **15th May 2024**

CONFERENCES & WORKSHOPS

6th Hands-On National Training Workshop on Advanced Bioinformatics and Recombinant DNA Technology 21/08/2024 – 22/08/2024

University of Central Punjab, Lahore

International Conference on Current Trends, Prospects, and Opportunities in Vaccine Research 28/02/2024–29/02/2024

Centre of Excellence in Molecular Biology, Lahore

6th International Conference on Applied Zoology (ICAZ-2023) 18/12/2023 – 19/12/2023

Government College University Faisalabad